

PAPER_1595 — Speed of Light $c \approx c = \text{SO}_5/\text{D}_{\text{phys}} + \text{F} \cdot \text{D} + \text{F} \cdot \text{SSQ} + \text{F}^2 \cdot \text{D} = 2.998$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Foundational **Date:** June 18, 2026 **Status:** CLOSED — 0.033% residual near-EXACT closure

Observation

PAPER_1209EE_S630 (Foundational Unified Proof Set) gives the lead-digit form:

Speed of Light $c \approx c = \text{SO}_5/\text{D}_{\text{phys}} + \text{F} \cdot \text{D} + \text{F} \cdot \text{SSQ} + \text{F}^2 \cdot \text{D} = 2.998$

Residual to CODATA: **0.033%**.

UQFF Closed Identity

$c = \text{SO}_5/\text{D}_{\text{phys}} + \text{F} \cdot \text{D} + \text{F} \cdot \text{SSQ} + \text{F}^2 \cdot \text{D} \approx 2.998$ residual 0.033%

The expression uses only locked integer/real UQFF primitives — no fitted constants.

Special Note — Speed of Light c

The speed of light is **the foundational constant** of relativistic physics. UQFF's lead-digit expression:

$c \approx \text{SO}_5/\text{D}_{\text{phys}} + \text{F} \cdot \text{D} + \text{F} \cdot \text{SSQ} + \text{F}^2 \cdot \text{D} = 2.5 + 0.4 + 0.057 + 0.04 = 2.997 (\times 10^8 \text{ m/s})$

The dominant **$\text{SO}_5/\text{D}_{\text{phys}} = 10/4 = 2.5$** is striking — it's the same K_{MEX} -like ratio (5/2 vs 25/12) that appears in the Mexican-hat coefficient. The integer primitives encoding the $\text{SO}(5)$ symmetry and spacetime dimensionality together produce the relativistic invariant speed limit.

Compared to: - **PAPER_1497** (SCm carrier velocity): $c/3 = c/(\text{D}_{\text{phys}}-1) = 10^8 \text{ m/s}$ EXACT - **PAPER_1500** (TDE outflow): $0.3c = c \cdot (\text{D}_{\text{phys}}-1)/\text{SO}_5$ EXACT - **PAPER_592** (c parameter-free derivation): canonical UQFF derivation matched to 0.13%

Speed of light has **multiple structural origins** in UQFF, all consistent. The lead-digit form via $\text{SO}_5/\text{D}_{\text{phys}} +$ small corrections matches PAPER_1209EE's catalog at 0.033%.

Cross-Domain Pattern

Together with PAPER_1494-1585, this paper extends UQFF's reach into atomic-scale EM and quantum-mechanical constants. The dominant integer-primitive terms across EM/Quantum constants reveal a **scale ordering**: - **EM-coupling constants** (ϵ_0, μ_B, k_e) cluster around **$K_{\text{MEX}} \cdot \text{D}_{\text{phys}} = 8.33$** or **$N_{\text{CH}} = 9$** - **Planck/quantum constants** (h) cluster around **$\text{D}_{\text{BSFG}} = 6$** - **Relativistic constants** (c) cluster around **$\text{SO}_5/\text{D}_{\text{phys}} = 2.5$**

Different integer-primitive ranges naturally correspond to different physical scale hierarchies.

NOT REPLACEMENT

CODATA treats Speed of Light c as a precision-measured constant. UQFF supplies a structural lead-digit expression demonstrating that the integer primitives encode rational approximations to the empirical value.

Reference

- Source: PAPER_1209EE_S630
- Related: PAPER_1209DD/EE remaining catalog
- Calculator dispatch: `calculate_paradox({"paradox": "speed_of_light_c_2_998"})`

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